American Journal of Germanic Linguistics & Literatures **9.1** (1997): 83–111.

Umlautless Residues in Germanic

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Evidence suggests that Germanic languages resisted the spread of historical umlaut processes. We propose that examples of such superficially varied umlautless residues all yield to a single coherent phonological account. Specifically, these vocalic assimilations show strong preferences for reducing more extreme differences in place of articulation between trigger and target while failing to assimilate articulatorily closer vowels, so that triggering /i, j/ first and most consistently mutated /a/ and last and least consistently mutated /u(:)/. This vocalic cline interacts with the consonantal material intervening between trigger and target, so that obstruent clusters (especially velar or labial) are prone to inhibit umlaut. In both cases, the smaller the phonetic difference between target and trigger, the less likely umlaut is to occur. While umlaut failure has long been consigned to the margins of theories of umlaut, we argue that it is crucial, providing a snapshot of how umlaut unfolded across western and north Germanic.*

"The real point at issue is the scope of the phonetic correspondence-classes and the significance of the residues. The neo-grammarians claimed that the results of study justified us . . . in seeking a complete analysis of the residues."

Bloomfield 1933:354

1. Introduction. While all Germanic languages save Gothic show various forms of the vowel harmony process commonly known as umlaut, much evidence indicates that all of these languages resisted the spread of umlaut in some ways. Umlautless residues, including

* We thank the following colleagues for discussions on this topic and comments on drafts of this paper and other help: two anonymous reviewers for this journal, Garry Davis, Stuart Davis, Greg Iverson, Monica Macaulay, Mary Niepokuj, Edgar Polomé, and Jolanda Vanderwal Taylor. An earlier version of this paper was presented at the Germanic Linguistics Annual Conference (GLAC) in Ann Arbor, April 20–22, 1995, where we benefited from discussion with numerous members of the audience. The usual disclaimers apply.

cases of the failure of nonprimary umlaut in Upper German—exemplified in 1 below—have been the subject of intense discussion for over a century, discussions in which phonological solutions have been out of fashion for decades.

(1) Umlautless forms (Schirmunski 1962:201–3)

	Dialect	Standard	Earlier form
a.	muck	Mücke	OHG mucka, cf. OS muggia
	šduk	Stück	OHG stucki
b.	khuxə	Küche	OHG kuchina
	lu:gə	Lüge	OHG lugin
c.	slupfə	schlüpfen	MHG slüpfen
	lupfə	lüpfen	Gmc. *luppian
d.	dro:^{n}mə	träumen	MHG tröumen
	ruəp	Rübe	OHG ruoba < *rôbjo

A procession of noted scholars since the 1960s have denied outright that sound change can account for such recalcitrant data, and they resort instead to purely analogical accounts. Building both phonological and comparative arguments, we propose that umlautless residues, while complex and variable, in fact reflect well-attested and comprehensible patterns of sound change rather than morphological realignment, and, moreover, they have clear parallels across Germanic. Even the apparent lexical irregularity at the margins of umlaut is consistent with data from other vowel harmony systems.

Phonologically, umlaut shows strong preferences for reducing the most extreme differences in place of articulation between trigger, or umlaut factor, and target, or affected vowel, while failing to assimilate articulatorily closer vowels, so that for *i*-umlaut, triggering /i, j/ first and consistently mutated /a/ and last but only inconsistently mutated /u/.

This varying effect by distance across the vowel space interacts further with consonants intervening between trigger and target, where umlaut fails with short /u/ when a geminate, especially velar, stop intervenes between target and trigger. Comparatively, these data fit closely with broader evidence from Germanic, including Netherlandic, Old English, and Old Norse. The basic principle underlying these superficially diverse phenomena is, again, simply that more similar groups are less likely to show assimilation. Languages around the world proffer parallels indicating that the phenomenon known as "umlaut failure" or "blocking" in Germanic is part of a well-attested (and thoroughly explicable) phonological process.

We aim then to place umlaut failure back into the realm of sound change and integrate it into an emerging broader school of thought on umlaut. The American structuralist theory of umlaut, pioneered by Twaddell (1938) and continued into the present decade by Penzl (1949, 1994), Antonsen (1969, etc.), and others, remains a widely accepted view, one that Keller (1978:160) calls "one of the finest achievements of American linguists." An analogical approach to umlaut failure becomes necessary under this traditional, Twaddellian analysis, which regards umlaut as monolithic and thus exceptionless. Iverson and Salmons (1996) argue that this approach fails on a variety of descriptive and other grounds, notably that the assumption of a single, simultaneous umlaut process in Old High German cannot possibly handle any umlaut blocking phenomena. Crucially for the present study, the Twaddellian view must explicitly deny any phonological regularity and is thus forced to favor analogical accounts over phonologically regular ones. Recognition of this dilemma, we suggest, was the basic motivation for works such as Penzl 1949 and Antonsen 1969. As Penzl puts it: "If a form without umlaut is not analogical, it can only go back to a form from which the *i*-sound was lost before the development of those allophonic variations that led to umlaut" (1949). This view is later sharpened by Antonsen (1966:120): "If a form without umlaut is not analogical, it can only go back to a form which already lacked /-i-/ in Proto-Germanic." These strong views seem to reflect an implicit understanding of the Achilles Heel of Twaddell's account, so that we in some sense draw inspiration from the insights of Penzl and Antonsen.

The attempt to justify analogical approaches to umlaut failure is founded on two points. First, Antonsen (1969) finds umlauted forms of all types, including /u/ with velar geminates, in Upper German. Given that umlaut failure is a strongly recessive feature, this is to be expected; the relevant point is that some dialects show umlaut blocking in the environments in question: If a single dialect shows a phonologically coherent distribution, we regard claims of analogy as untenable. In fact, many dialects show the regular patterns we describe, even one thousand years after umlaut ceased to be phonologically active, as we will see below. The second objection is that no rule can be defined and motivated (Lüssy 1974). Yet, the basic regularities in this dataset are remarkable. That umlaut failure occurs only with high back vowels but never mid or low vowels in Old High German alone is fatal to the analogical view, even before we consider evidence from other Germanic languages or intervening consonants.[1] In moving umlaut failure back into the discussion, this paper contributes to the growing body of opinion that umlaut was hardly the monolithic change posited by Twaddell, but instead a process that can only be seen coherently in terms of long, gradual evolution (Buccini 1992, Iverson, Davis, and Salmons 1994, Fertig 1996, Iverson and Salmons 1996).[2]

In the rest of this paper, we first establish the central importance of umlaut failure in a broader theory of umlaut and then develop a principle for which vowels are most and least affected by umlaut, which we adapt to cover intervening consonant blocking as well. We close by dealing briefly with minor residual irregularities in umlaut and other vowel harmony systems.

2. The Importance of Exceptions to Umlaut. An adequate explanatory account of umlaut processes in Germanic must go beyond mere description of the conditioning environments and the effects they exert on preceding vowels. Simple description of umlaut processes leaves key questions unanswered. It is, for example, crucial to our understanding of umlaut to review precisely which sequences of vowel-intervening consonant(s)-vowel tend to trigger umlaut processes in the various dialects of Germanic and which sequences rarely or never induce such changes. The investigator must then

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[1] Labov has recently noted about unprincipled claims of lexical diffusion that "by accepting the word as the fundamental unit of change, the investigator is less likely to search for the deeper phonetic regularities that may underlie the process of change" (1994:450). This, mutatis mutandis, applies to analogical arguments, where proponents have declined to pursue the search for regularity far enough.

[2] Another aspect of this emerging view is that phonological solutions have been too often given short shrift in understanding umlaut. For example, Salmons (1994) argues that the connection between umlaut and plurality in Old High German is ultimately far more directly phonologically driven than many recent theories, especially natural morphology, have seen it as being.

provide a principled phonetic explanation of exactly why umlaut processes repeatedly arise cross-dialectally in certain environments while other sequences appear to impede umlaut and still other VC_0V sequences virtually never participate in umlaut. Although extant data may never suffice for a seamless account of umlaut in all its permutations in the dialects of Germanic, it is safe to say that the facts at our disposal have not been fully exploited. Specifically, too little attention has been paid the exceptions to umlaut processes. In the following paragraphs we will argue that it is precisely these jagged edges of umlaut rules that shed the most light on the kinds of conditions that tend to trigger umlaut and help to explain the subsequent generalization of umlaut to other environments.

Neogrammarian handbooks typically provide rather complete accounts of umlaut processes, prominently featuring a description of the conditioning environment and the resulting vocalic mutation. Major exceptions to the umlaut rules are also generally recorded in subsequent paragraphs whereas more idiosyncratic exceptions tend to be buried in the footnotes. More recent, feature-based accounts in the generative paradigm have focused on capturing umlaut processes in their most general form, while exceptions, which interfere with the generality of a rule, are treated cursorily or ignored altogether (beginning with King 1969, further developed by Lass and Anderson 1975, Barrack 1975, Voyles 1991, 1992, and many others). These approaches succeed in providing a rudimentary account of a basic umlaut rule and admirably capture the fact that umlaut processes constitute assimilation of feature(s) of the conditioning vowels by the umlauted vowel segment:

(2) Secondary umlaut in German

$$\underset{[+\text{ back}]}{\text{V}} \longrightarrow \underset{[-\text{ back}]}{\text{V}} \;/\_\_\; \text{C}_0 \; \underset{\substack{[-\text{ back}]\\ [+\text{ high}]}}{\text{V}}$$

This type of rule is not, however, even descriptively adequate if we include among our data phonologically well-defined blocking environments. Further, it cannot explain why the conditioning factor for secondary umlaut must be specified for height as well as for frontness, nor does it explain why the process fails to affect reflexes of

OHG *u*, *û* and *ou* in some Upper German dialects when certain consonants intervene between root vowel and the umlaut factor.

If it seems counterintuitive to claim that restrictions on umlaut can tell us much about the origin and spread of the process, one must realize that early textual evidence reflecting incomplete umlaut developments as well as more modern dialectal data showing incomplete operation of umlaut processes provide us with valuable "snapshots" of the developmental stages of umlaut. Old High German texts, for example, reflect German *i*-umlaut in its infancy. Given the fact that Old High German "primary umlaut" is limited to the fronting and raising of WGmc. */a/ in certain environments, we can view it as a paradigm example of a triggering environment for umlaut. On the other hand, the limited failure of "secondary umlaut" of OHG /u/, /û/, /ou/ in Upper German dialects of Middle High German reflects a late stage in the umlaut process in fossilized form, with the unumlauted reflexes of OHG /u/, /û/, /ou/ representing the vowels which conform least to the original trigger environment for OHG/MHG *i*-umlaut. By comparing various incomplete umlaut processes, we can, therefore, potentially arrive at a basic typology of umlaut and ascertain what conditions favor the development of umlaut-type assimilation and which environments tend to hinder umlaut.

Howell (1983) and Van Coetsem and Buccini (1988) make clear that umlaut processes are crucially affected both by the phonetic relationship between the umlaut factor and the preceding vowel and by the intervening consonant(s):

(3) V C V

Obviously only a limited subset of possible vocalic sequences in Germanic result in umlaut phenomena. This set of umlaut environments is in turn potentially affected by intervening consonants. An explanatory account of umlaut must deal with both, but the vocalic relationship is of primary importance.

3. Degrees of Difference: Trigger Environment for Umlaut. Umlaut processes in Germanic present a broad range of vocalic mutations including raising, lowering, fronting, backing, rounding, diphthongization, and combinations thereof. The uncontroversial unifying characteristics in this diversity are: 1) umlaut is an assimilatory process, and 2) the vowel-to-vowel assimilation is uniformly regressive in Germanic. Less clear is why umlaut affects only certain sequences of vowels. Even a partial answer to this question would represent a contribution to our understanding of the genesis and generalization of umlaut.

Any assimilation necessarily requires as input two phonetically different segments. This simple principle represents a point of departure for our discussion of the relationship between trigger and target in Germanic. In what follows we argue that it is the *degree* of difference between two vowels in height and frontness that determines the likelihood of their participation in an umlaut process. Those sequences that are most different with respect to tongue position will be most susceptible to umlaut and, conversely, those segments which are least different phonetically tend to undergo umlaut chronologically later or not at all. This principle, as an anonymous reader points out, follows naturally from Vennemann's (1988) "Diachronic Maxim," whereby change along a given parameter will always improve the worst structures first and the least offensive structures last.

In a broad theoretical discussion of general characteristics of feature spreading in vowel harmony languages, Cole and Kisseberth (1994) introduce two basic principles which "drive the existence" and, we would argue, the genesis of the spreading of vocalic features:

1) Perceptibility: Features should be perceptible, that is, "features serve to mark contrast, and must be perceptible to fulfill their function."
2) Articulator stability: Minimize changes from the neutral, steady state of the articulators, that is, "features are implemented by an articulatory system which imposes its own constraints on the realization of features" (Cole and Kisseberth 1994).

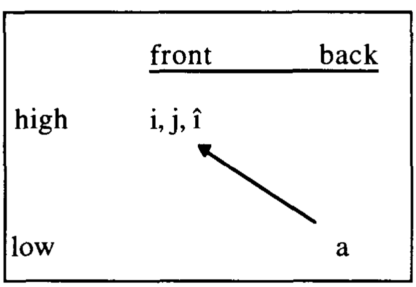
These two principles are ideally met when a feature is realized over more than one segment, which Cole and Kisseberth express in their Principle of Extension:

3) Extension: Extend a feature over longer stretches of sound in order to maximize perceptibility and articulator stability.[3]

All instances of umlaut in Germanic satisfy the Principle of Extension 1) by maximizing perception of a feature or features of an umlaut factor by extending it to a preceding vowel and 2) by increasing articulator stability through reduction of articulatory distance between umlaut factor and umlauted vowel.

While both perceptibility and articulator stability are enhanced in the output of Germanic umlaut processes, evidence provided by incomplete umlaut rules indicates that articulator stability plays a decisive role in the chronology of umlaut. In the development of *i*-umlaut in German, we see that the first vowel affected is the short vowel most phonetically different from the conditioning factors, namely /a/:

(4) Primary umlaut in German



[3] Cole and Kisseberth (1994:4) recognize the fact that there is an "inherent tension between the Principle of Extension and the functional role of a feature to mark contrasts, since when a F[eature]-domain is extended, the possibility of contrast between [F] and its absence is eliminated in all positions within the extended domains." But they also simply assume that languages resolve this tension in different ways. In Germanic, for example, the extension of frontness and/or height of umlaut factors to umlauted vowels most often does not result in a complete assimilation to the umlaut factor, but rather in umlauted vowels which share features of the original vowel and the umlaut factor. In German, *i*-umlaut of WGmc. */o/, */u/ to MHG /ö/, /ü/ results in vowels that reflect the roundness and height of the original vowels and the frontness of the umlaut factors */i/, */j/, */î/.

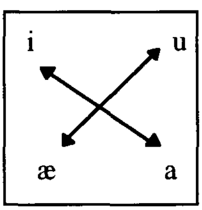
Examples: OHG *gast* ~ *gesti* 'guest/guests'; OHG *faru* ~ *feris* ~ *ferit* '(I) go, (you) go, (s/he) goes' (Braune and Eggers 1987:26–29)

Primary umlaut, which must precede secondary umlaut chronologically,[4] illustrates the trigger environment for umlaut processes in Germanic (within the limits of traditional vowel features to capture the three-way height distinction):

(4.1) Trigger environment for umlaut (in features)

target	intervening C[5]	trigger
|		|
V	C_0	V
[α high]		[$-\alpha$ high]
[β back]		[$-\beta$ back]

(4.2) Trigger environment for umlaut (schematically represented)



[4] See Iverson and Salmons 1996. Arguments in favor of placing primary umlaut chronologically earlier than secondary umlaut include: 1) primary umlaut is the only process to affect the entire continental West Germanic area. Secondary umlaut, including umlaut of */a/ in blocking environments, does not occur in western Dutch (examples given below), 2) Nonumlaut of WGmc. */a/ in "blocking environments" dialectally in Old High German implies later onset of at least this part of what is commonly called secondary umlaut, and 3) primary umlaut and secondary umlaut result in fundamentally different vocalic mutations—it is, in fact, difficult if not impossible to state the results of the two processes in one phonological rule. Primary umlaut causes the fronting AND raising of WGmc. */a/, while secondary umlaut causes only the fronting of the affected vowels, a point going back at least to Vennemann (1972:882).

[5] It is worth noting that umlaut is least constrained when no consonant intervenes, cf. the great extent of the so-called *w*-breaking in Old English and the greater range of *u*-umlaut in Old Norse when umlauting *u* directly follows a root vowel (Noreen 1970:69–72).

Umlaut processes in Germanic indicate that the umlaut effect generalizes from this optimal trigger environment to environments where phonetic differences between conditioning factor and target vowel are less dramatic. In those instances when umlaut processes do not reach their full potential extent, the failure occurs as the umlaut factor, or trigger, and the target vowel become more similar phonetically. For the various types of umlaut in Germanic we therefore posit the origin and direction of spread depicted in figures 5.1–5.4.

Umlaut has spread out in waves, stopping at points that show its historical unfolding. On opposite ends of the West Germanic dialect continuum we find interesting examples of partially completed *i*-umlaut processes that illustrate the significance of the trigger environment for umlaut depicted in figures 4.1–4.2 above. In coastal Dutch, *i*-umlaut affects only WGmc. */a/ with complete regularity, the environment most susceptible to *i*-umlaut according to the tendential cline of umlaut resistance developed here. Only intervening *-ht-* clusters block the umlaut effect, a distribution reminiscent of that found for primary umlaut in Old High German. Distribution of the *i*-umlaut of WGmc. short **u* is, as Goossens (1980:34) states, "chaotisch," often with unumlauted forms where an original *i*-umlaut factor was present and umlauted vowels in words with no original umlaut factor. The presence of unumlauted forms with an original *i*-umlaut factor points to a more limited distribution of *i*-umlaut of **u* in these coastal dialects. In the western dialects all long vowels remain unaffected by *i*-umlaut—that is, what we traditionally call "secondary umlaut" in German does not affect coastal Dutch.[6] In these dialects, the umlaut process has been arrested at an early date (cf. Buccini 1992), regularly affecting only environments where target and trigger are phonetically most highly differentiated.

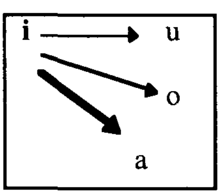
It is worth noting that the areal distribution of *i*-umlaut of the long vowels in Dutch is predicted by the degree of articulatory difference between target and trigger. WGmc. **â* shows *i*-umlaut in the entire southern part of the Dutch linguistic area except for a small West Flemish coastal enclave. In Holland, unumlauted forms are found in all of North and South Holland as well as in much of the province of

[6] The situation among the short vowels is obscured particularly by the so-called "spontaneous palatalization" of old /u/.

Utrecht to the east (Weijnen 1991:262). Reflexes of umlauted WGmc. **ô* and **au* are, except in a small part of Utrecht, found well to the east of the isogloss for umlaut of **â* and at no point reach the coast (Goossens 1977:91). The extent of *i*-umlaut of WGmc. long **û* can not be evaluated because the spontaneous palatalization of all reflexes of **û* masks the distribution of umlauted forms. We therefore find that in Dutch, those long vowels that show a greater phonetic difference from the umlaut factor prove more susceptible to the *i*-umlaut process as reflected by their more extensive areal distribution.[7]

In Bavarian and Alemannic, on the other hand, we encounter a much more extensive operation of *i*-umlaut. Nonetheless, many dialects show failure of *i*-umlaut when the target contains [u], the vowel predicted to be least susceptible to *i*-umlaut by our account. In these dialects, *i*-umlaut (i.e., "primary" and "secondary" umlaut) has affected long and short low and mid vowels, but the fronting of high back vowels is more restricted and, as is so often the case in incomplete umlaut processes in Germanic, limited by the nature of the intervening consonant(s). Just as Dutch shows *i*-umlaut only when target and trigger are maximally different with respect to height and frontness, i-umlaut fails in Upper German dialects *only* when the difference between target and trigger is least:

(5.1) *i*-umlaut (trigger WGmc. */i/, */j/, */î/)



A) Coastal Dutch:
1) *i*-umlaut: MDu. *gewelt* 'power', *denken* 'think', *best* 'best', *bed* 'bed', *mensche* 'person', *tellen* 'count', *temmen* 'be fitting' (< WGmc. **giwaldi, *þankjan, *batist-, *badi, *manisk-, *talljan, *tammjan*) (Van Loey 1976:14–15)

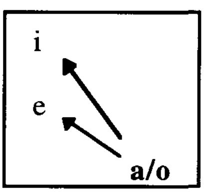
[7] For an exhaustive set of examples of umlauted versus unumlauted forms in Middle Dutch, see Van Loey (1937:28–94).

2) no *i*-umlaut: MDu. *machtig* 'mighty', *kaas* 'cheese', *horen* 'hear', *schoon* 'beautiful', *groeten* 'greet' (vs. German *mächtig, Käse, hören, schön, grüßen)*

B) Upper German:
Failure of *i*-umlaut in Middle High German
1) OHG /û/: UG *rûmen* 'räumen', *sûmen* 'säumen'
2) OHG /uo/: UG *uoben* 'üben', *suochen* 'suchen'
3) OHG /ou/: *gelouben* 'glauben', *touben* 'betäuben', *houbet* 'Haupt', *zoubern* 'zaubern' (Paul, Wiehl, Grosse 1989:67)

In West Saxon dialects of Old English the limited extent of velar umlaut provides a phonetic mirror image of *i*-umlaut depicted above. West Saxon texts show a very limited distribution of *o/a*-umlaut: only high front /i/ is affected by the low back trigger, and, once again, the process is sensitive to the nature of the intervening consonantism. *o/a*-umlaut of /i/ occurs only before a single liquid or a non-geminate labial:

(5.2) West Saxon *o/a*-umlaut

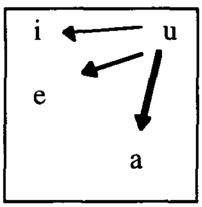


Examples: Pre-OE **i* > WS *eo* /_ C_1 (liquid/labial) *o*: *cleopoda* 'called', *teoloda* 'worked', *bîleofa* 'food', *seofoða* 'seventh', *heora* 'their' (Brunner 1965:87); Pre-OE **e* > 'nicht streng' WS *eo* /_ *(l, r)o*: *steola* 'handle', *beoran* 'bear' (Brunner 1965:86).

Some West Saxon texts show a limited *o/a*-umlaut of *e* when only a liquid intervenes. Once again, the umlaut effect weakens as target and trigger become more phonetically similar.

The distribution of the Old Norse *u*-umlaut and *u*-breaking processes also serve to illustrate the strong mutating influence exerted by the umlaut trigger environment, while the umlaut effect becomes increasingly restricted as target and trigger share height features. The low vowel *a* therefore undergoes unrestricted *u*-umlaut, short mid *e* is diphthongized (so-called 'breaking'), but original high short *i* is rounded only under a very restricted set of circumstances: when immediately preceded by a labial consonant or when the umlaut factor developed from an original **w*.

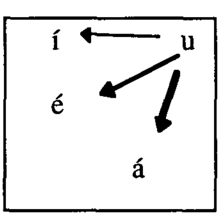
(5.3.1) *u*-umlaut/*u*-breaking of short vowels in Old Norse



1) *u*-umlaut of **a* > ǫ *mǫgr* 'son' (cf. Go. *magus*) *sǫk* 'affair'; *lǫnd* 'lands' (< **landu*)
2) *u*-breaking of *e > io: *hiotr* 'deer'; *hiorr* 'sword; (cf. Go. *haírus*); *ioforr* 'boar'
3) *u*-umlaut of **i* > *y* only when umlaut factor develops from **w* or when target vowel is preceded by a labial: *nykr* 'water-sprite' (< **nikuz* < **nikwiz*), *myklom* dat. pl. 'large' (Noreen 1970:69–79, 87)

Long vowels in Old Norse show a similar but even more restricted distribution of *u*-umlaut, with low long *á* undergoing backing and labialization regularly but with long mid *é* and long high *í* showing rounding only when the trigger *u* directly followed:

(5.3.2) *u*-umlaut/*u*-breaking of long vowels in Old Norse



1) *u*-umlaut of **á* > *ǫ́*: *ǫ́tom* 'we ate', *skǫ́l* 'bowl'
2) *u*-umlaut of **é*, **í* > *ǿ*, *ý* only when *u* follows directly: *fǿ* 'cattle' (< **fēu* < **fehu*), *bý* 'bees' (<**bīu*)

In all of the umlaut processes illustrated above, greater phonetic similarity between the trigger and target vowels correlates with a weaker umlaut effect. This kind of distribution for umlaut processes is consistent with and provides additional evidence for Cole and Kisseberth's Principle of Articulator Stability. Clearly, all umlaut processes in Germanic ultimately work to "minimize changes from the neutral, steady state of the articulators" and the evidence presented here indicates that umlaut processes in fact originate in and spread from environments where the original, pre-umlaut transition from target to trigger vowel shows a maximal change "from the neutral, steady state of the articulator." In target-trigger sequences where the transition from vowel to vowel results in less articulatory instability, the umlaut process is more weakly motivated phonetically, predictably resulting in a more constrained set of umlaut environments and/or chronologically later onset of the umlaut effect.

While the phonetic relationship between target and trigger vowel provides the basic impetus for umlaut processes, it is clear from the distribution of umlaut in Germanic that consonants intervening between the two vowels can have a crucial limiting effect on the operation of umlaut. In every umlaut process illustrated in figures 5.1–5.3, the nature of the intervening consonantism is critical in defining the umlaut environment when the target and trigger conform less completely to the Trigger Environment for umlaut. Our account, focusing on similarity/difference between target and trigger (with intervening consonants) can therefore account for the fact that failure of "secondary umlaut" in Upper German is found only when the target vowel is, like the trigger, high—but the fact that this failure of umlaut is also affected by the intervening consonantism must be explained.

4. Intervening Consonants. That intervening consonants contribute, often decisively, to umlaut failure is hardly a new notion. Blocking of umlaut by velar geminates following /u/ has been recognized since at least von Bahder (1890, cf. Lüssy 1974:81–83) and is so striking that Schirmunski (1962:201) calls that subtype "regular" in Upper

German.[8] For example, Vetsch describes Appenzeller dialects as consistently lacking umlaut of MHG short /u/ before geminates (1910:70–72), lacking it in /u:/ before geminate /m/ (1910:77) and so on.[9] While the most systematic and geographically widespread umlaut failure in Old High German occurs with velar geminates, it is also well attested with labials, less commonly with coronals, and sporadically elsewhere, as seen in the data in Lüssy 1974 and other sources. Recall that greater differences are more likely to be assimilated earlier in the unfolding of umlaut, while lesser differences are the last and least likely to be assimilated. Degree of phonetic differentiation covers both vocalic and consonantal aspects of blocking: the more intervening consonants have in common with target vowels, the likelier umlaut failure.

If we characterize blocking environments in place features, we see that not only the vowels, but the entire sequence in the prototypical blocking environment, forms of the "ukxi" type, shares the aperture feature [+high] and all segments save the trigger share [+dorsal], the maximum possible amount of place sharing.[10] Exactly this sequence is predicted by our principle to be least likely to harmonize. With intervening labial geminates, the target and consonants share [+labial], one place feature (differing from the trigger vowel), while coronal geminates share nothing. This matches with the attested patterns, shown in 6:

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[8] It has been noted that the dialects showing the most consistent blocking are not the most conservative Swiss German dialects, as an argument against these residues representing phonological failure of umlaut. Of course, dialects showing umlaut failure are all extremely conservative, including even nonreduction and loss of final vowels, key to umlaut (cf. Schirmunski 1962:162–64). At any rate, retention of one archaic feature is not limited, in principle or in practice, solely to the most archaic varieties, but we find rather complex patterns of archaism and innovation, in German dialects and elsewhere.

[9] Vetsch does think that umlaut has spread into some lexical items analogically. If there is analogical change after it ceased to work phonologically, it can under our analysis only be toward, not away from umlaut, but we note also the possibility of borrowing of umlauted forms into the dialect.

[10] We assume here, as has much other recent work on Germanic historical phonology, Clements's model of feature geometry, with a vowel place node, rather than treating all vowels as dorsal.

(6) Geminate blocking

		Most common	⟷	Least common
		-ukxi	-upfi	-utsi
Aperture	[±high]	++++	+−−+	+−−+
Place	[±coronal]	−−−+	−−−+	−+++
	[±labial]	+−−−	+++−	+−−−
	[±dorsal]	+++−	+−−−	+−−−

(6a) Alemannic examples of blocking by place of intervening geminate, from Lüssy 1974

-kx- *bucke* 'bücken', *drucke*, *hucke* 'sich ducken, hinken', *jucke* 'springen', *lucke* 'locken', *nucke* 'einschlafen'

-pf- *gupfe* 'emporschnellen', *hupfe* 'hüpfen', *lupfe* 'heben', *mupfe* 'einen Stoß geben'

-ts- *butze* 'putzen, kastrieren', *gutze* 'überschießen', *hutze* 'auf-springen', *nutze* 'nützen', *stutze* 'stehenbleiben'

This cline fits still more precisely if we avail ourselves of a recent proposal about place features based on broad cross-linguistic data from place assimilations. Using data from various Native American languages, Korean, English, and other languages, Rice (1994) points to a consistent but asymmetric pattern whereby coronals assimilate to labials and velars, labials to velars but not to coronals, while velars do not assimilate. This indicates that phonologically, velars have more place structure than labials and labials more than coronals. Rice captures this generalization formally by positing for place features a hierarchical structure like 7a rather than a flat one (7b) or the more directly articulatorily oriented proposal of Clements and Hume (1994) (7c).[11]

[11] Note that this configuration is in some sense consistent with the notion of "articulator stability" discussed above, since less marked place requires less departure from the neutral position.

(7) Models of place features

a.

Place

Peripheral (Coronal)

Dorsal (Labial)

[Parentheses = unmarked features]

b.

Place

Labial Coronal Dorsal

c.

Place

Labial Lingual

Coronal Dorsal

Crudely put, the best blocking environment is the one with the greatest amount of place structure: a dorsal linked to two timing slots.[12] Rice's proposal mandates that since dorsals have the most structure, they serve as triggers for assimilation, that is, allow their structure to spread, but not as targets, since structure cannot spread into their existing structure. This theory predicts that dorsals should be the most likely segments to block a distant place assimilation, then labials, with coronals most likely to be transparent.[13]

Still, it is not place features of intervening consonants alone that lead to consistent blocking but place specifically of geminates. The featural configurations assumed make blocking here feasible and even intuitively appealing and consistent with the proposed account, but we

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[12] An anonymous reader notes that long vowels are more resistant to change than short ones because they have "more structural stability in all regards," pointing out a possible parallel to consonantal blocking here.

[13] In fact, this place structure also bears directly on the geographical spread of the second or High German consonant shift in somewhat similar ways, as argued by Davis, Iverson, and Salmons (manuscript).

have not yet accounted for this relationship in terms of modern phonological theory, a task to which we now turn. Consonantal blocking of vowel harmony and umlaut processes has been the source of considerable theoretical maneuvering, even head-scratching. Most feature geometries put vowel and consonant place on different tiers, predicting that consonants will generally be transparent. When consonantal material interacts with harmony, all consonants usually block. Beyond this, the well-known "laryngeal transparency"—harmony across [h] and [ʔ] in languages where consonants otherwise block—is easily enough explicable, since laryngeals lack (supra-laryngeal) place specification. But where some consonants do and others do not block harmony, things get complicated. In consonants that contain "secondary place" specifications, like labiality on a velar stop in [k^w], these secondary place features are treated, under the theoretical assumptions of those like Clements, as vowel place and can, by virtue of this, block harmony (an idea already exploited for Old High German by Iverson, Davis, and Salmons 1994).[14] Geminate blocking does not obviously fit into any of these types, and requires another approach.

We see three plausible phonological accounts of this problem. We will describe the first in some detail and note the other two only briefly. First, we suggest that the Autosegmental Linking Condition is responsible for blocking here (just as has been argued for Korean (Iverson and Lee, forthcoming) and OHG *-hC*-blocking (Iverson, Davis, and Salmons 1994). This principle says that "in order to change the feature content of a segment [A] every skeletal slot linked to [A] must satisfy the rule" (Hayes 1986, this formulation from Kenstowicz 1994:413). Where blocking occurs, it is because of shared place features. Geminates are by definition multiply linked and thus inalterable. Again, the more features shared and the more marked those features are, the more likely blocking becomes.

Variability in blocking can derive, as Greg Iverson suggests to us, from two possible configurations, one with feature-sharing and one without. That is, sequences of the *-ukxi* type can be represented either

[14] This should exhaust the possible types of blocking, pace Clements (1991), but Casali (1993) finds that Clements-type geometries cannot handle Nawuri labial blocking of roundness harmony, so these matters remain controversial.

as feature sharing or having two independent instantiations of place features. The former results in blocking, while the latter does not.

A second way that feature geometry can help motivate this blocking pattern, is specifically by reference to the affricate status of the blocking consonants. In works on the structure of "complex segments," Steriade (1993 and elsewhere, see also Kenstowicz 1994:503–6) has proposed that stops consist of two distinct phases, closure and release, with affricates characterized as stops with fricative release rather than full or nonrelease. As Kenstowicz indicates, this "release position is a natural docking site for aspiration and glottalization and perhaps secondary place specifications such as labialization and palatalization" (1994:503). In short, affricates can be seen as having secondary place as a way of capturing the fricative release. In this case, where the affricates are ambisyllabic or disegmental, the consonants could still share this structure. Incorporating Rice's internal organization of the place node into this scheme predicts exactly the tendential hierarchy we find among the old geminates: velars contain the most place structure and are thus most likely to block and so forth. On Steriade's view, then, geminate place structures are subordinate to C-place features, offering a close parallel to blocking by segments with secondary place features.[15]

Third, Stuart Davis (1994) treats a case of geminate umlaut blocking in Korean and argues that umlaut in Korean and cross-linguistically is sensitive to syllable weight. Geminate codas give the target syllable enough weight to make it umlaut-resistant. This view, while we will not pursue it here, is enticing given the weight sensitivity of Germanic phonology in general, as familiar from such classic works as Murray and Vennemann (1983). Primary umlaut itself was, after all, weight sensitive, in its inherent limitation to a short vowel.

Blocking is, however, not entirely limited to geminates, but occurs with *-mb-* clusters as in *umbi* 'around' (cf. umlauted variants farther north), etc., if only sporadically. Kenstowicz (1994:422–23) in fact suggests in a discussion of Kolami vowel copying that geminate inalterability may be extendible to clusters sharing place of articulation, coincidentally with nasal + oral stop clusters. Nonetheless,

[15] While we do not go so far as to connect vowel height directly to Steriade's stop aperture, it is at least interesting to note that it is stops with closer release aperture—i.e., affricates—that block.

we have additional reason to understand place-sharing clusters as sporadic blockers. Nongeminate blocking still most often involves two intervening consonants, so that the target syllable generally has a coda. Clements (1991) has proposed a unified set of features common to vowel and consonant place and demonstrated that "there is pressure for agreement in place features across tiers" (1991:16) in CV sequences.[16] Clements argues for preferential sharing of place specifications, such as sequences of [+labial] and those of [+dorsal], viz. labial consonant + back rounded vowel and velar consonant plus back rounded vowel. An obstruent marked for place in the coda of the target syllable could inhibit spread when place features are shared between nucleus and coda of the target syllable. Existing target rhymes of *-uk* or *-up* represent the preferred, feature-sharing structure, which is (tendentially) not disturbed by umlaut.[17]

Geometrically, this preference is realized as features shared on the coda consonant as subordinates of C-place, resulting in blocking, and we have variability because this preferred sharing is not pervasive but competes with independent instantiations of the features on nucleus and coda.[18]

This hypothesis that coda obstruents are preferred when they share place features with the nucleus is, in fact, dramatically confirmed by comparative Germanic evidence. In seventeenth-century English, the unrounding of /ʊ/ to [ʌ] is inhibited by a preceding labial—as in *bull, pull, full, wolf*—or following velar—*cook, look, rook* (Jespersen 1909, Clements 1990, Milroy 1992). That is, labiality is retained

[16] While monoplanal line-crossing is of course still prohibited, this introduces the possibility of line-crossing *across* planes.

[17] Note that this point—a preference for codas and nuclei to share their place features—also gives support to blocking across geminates as well, since velar and labial geminates represent by definition place-sharing rhymes of target syllables.

[18] This may have more direct prosodic motivation as well. Codas inevitably suffer restrictions on what features and segments they can bear (for example, Vennemann 1988, Goldsmith 1990, and Clements 1991), so that this is a conservative hypothesis on theoretical grounds, where feature sharing between nucleus and coda is an intuitively attractive way of reducing distinctiveness of the coda.

between onset and nucleus, dorsality between nucleus and coda. Jespersen's case parallels our (nongeminate) blocking data perfectly: both show resistance to an otherwise regular loss of dorsality on vowels when the rhyme shares [+dors] and [+hi]. Yet more pronounced instances of place sharing between nucleus and coda can be observed in Germanic, though, both modern and ancient. Milroy (1992) reports that Belfast [æ] is raised to [ɛ] before velar consonants in monosyllable codas, as observed in the dialect of Madison, Wisconsin as well.[19] This kind of example shows particularly strong preference for place sharing, since C-place features of codas come to be shared with nuclei rather than as in our situation, where sharing simply prevents changes away from shared structure.

"Back" or *u*-umlaut in Old English and Old Frisian shows a related inclination at an earlier date. Old Frisian velar obstruents followed by /w/ or /u/ change /i/ into /iu/, that is, backness is regressively assimilated into the nucleus of a preceding syllable (Sjölin 1969, etc.), as in 8:

(8)	Gmc.	Old Frisian	Other Germanic languages
	*-iŋkw-	diunk- 'dark'	ON døkkr
	*-ikkw-	thiukke 'thick, fat'	OS thikki
	*-iŋgw-	siunga 'sing'	ON syngva
	*-igu-	niugun 'nine'	OS nigun

Dorsality, shared by trigger and all intervening material, comes to be shared by the target vowel.

To summarize this section, the principle governing which vowels trigger and are targeted by umlaut governs consonant blocking as well: the more place features the target shares with intervening consonants, the more likely umlaut failure becomes. Given that various West Germanic languages spread place features to target vowels in this environment, and given attested resistance to losing identical features in sound change, it is not controversial to suggest that sequences with a target vowel sharing the appropriate place features are inclined to resist fronting processes at least sporadically.

[19] Note that height gets assimilated, not backness, in these two possible first steps toward a harmony system. Buccini (1992) establishes a typological universal that height harmony necessarily precedes frontness harmony.

This produces a viable phonological account of blocking and one intimately connected to primary umlaut blocking in the analysis of Iverson, Davis, and Salmons (1994), giving a unified account of umlaut blocking from beginning to end, with the Autosegmental Linking Condition playing a role throughout the long process.

5. Remaining Apparent Irregularity in Umlaut Failure. The fact that the residues in the dataset are so few and so variable has suggested to many that they do not reflect regular sound change. Two processes have been proposed to account for lexical irregularities, analogy, and lexical diffusion. The first has already been shown unlikely above and, while some have hinted about lexical diffusion of umlaut (e.g., Hamans 1985), we will not pursue the second here. Our analysis rests instead on showing the basic regularity of umlaut failure in and as sound change, but a handful of problematic forms remain, leaving our rules less than airtight. We are still obligated to show that these are consistent with the analysis developed here, phonologically, historically, and perhaps even socially.

With distant assimilations, such wrinkles are remarkably common and attested in active phonological processes and recent past changes, some showing parallels to the feature-sharing arguments advanced here. Some Korean dialects, for instance, show umlaut and umlaut blocking patterns often like those of OHG primary umlaut. According to Iverson and Lee (forthcoming), a phonologically active umlaut process in the Korean dialects of Kyungsang and Cheolla is "sociolinguistically variable and subject to considerable lexical exceptionality." For Nawuri, Casali (1993:3–6) reports blocking of roundness harmony by intervening labial consonants, but this blocking is "optional and speech-rate dependent." Nawuri centralization harmony "sometimes fails to occur . . . when one or both of the flanking consonants is a palatal." In both of these systems, as in Germanic, place sharing seems to be central to harmony and failure, while both show lexical exceptionality. Other, if less precise parallels in lexical exceptionality in harmony systems are found around the world. Kenstowicz (1994:354–55) reports a handful of cases of failure of ATR harmony in Akan. Clements and Sezer (1982:249–50) are able to account for most disharmonic forms in

Turkish, yet still find themselves with a residue, instances where harmony processes can be "optional."[20]

In short, lexical irregularity in umlaut failure—the justification for considering analogy—is consistent with vowel harmony processes around the world, in settings where "analogy" cannot be seriously appealed to.[21] If this limited irregularity presents a problem—and we do not see it as a serious one—it is a problem for theories of vowel harmony generally, not a problem for the Upper German case in particular.[22] If anything, the regularity in attested Upper German dialects is astonishing, given the antiquity of umlaut itself. That is, if morphological realignments had been busily undoing phonological coherent patterns of umlaut for over a millennium, we should find far less phonological regularity than we find in more active harmony processes, perhaps even the complete obliteration of phonological patterns by analogy. Instead we find roughly the same patterns of regularity and exceptions in Upper German dialect umlaut and active harmony systems.

While better understanding of prosody may illuminate exceptions, a few may ultimately have a social explanation. "Late stage

[20] See also Kirchner 1993.

[21] We leave aside here later and clearly analogical processes, such as the elimination of umlaut from the singular of some nominal paradigms, such as the dative and genitive singular forms of OHG masculine *n*-stem nouns. Note that this process is substantially different from the analogical spread of umlaut to new lexical items.

[22] We even leave aside the possibility that more detailed phonetic conditioning is at work than has hitherto been uncovered, just as Labov (1994:426–28, 444–48) shows subtler distant conditioning for the Atayalic labial/velar shift, previously seen as lexical diffusion. For instance, in the limited set of Old High German verbs with medial labial affricates, i.e., medial consonants ultimately derived from geminates (following the data in Lüssy 1974:105, 109–10), some show competition between umlauted and umlautless forms (*güpfe/gupfe; hüpfe/hupfe; lüpfe/lupfe; müpfe/mupfe; rüpfe/rupfe*), while another shows only umlautless forms (*schlupfe, schnupfe, schupfe, strupfe, stupfe, verstumpfe, supfe, zupfe*). The first set contains an array of different but always simplex onsets, while the second set always contains sibilants and seven of eight contain complex onsets. A similar, if weaker pattern of this type exists for the fem. *-jō(n)* stems (1974:89).

correction" has become an established part of studies of sound change in progress and it fits well with the pattern of lexical exceptions we find here. In late stage correction, lexical exceptions to otherwise regular sound change reflect borrowings from "higher" varieties that have not undergone the sound change in question. The change in question is typical of "change from below," that is, changes originating in the vernacular and generally reflecting the "operation of internal, linguistic factors" (Labov 1994:78). Speakers are not conscious of such change until it reaches its last stages and when they do become aware of it, new forms can be stigmatized, motivating reintroduction of a few unchanged forms. Labov (1994:453–54) reanalyzes a number of putative cases of lexical diffusion as "late stage correction," including the well-known Chaozhou tone split. While we lack the necessary social information to prove this case, we note its plausibility. The handful of Middle High German forms still blocking with -LC- clusters, for example, would fit that pattern particularly nicely: *arte* 'sort, kind', *einfalte* 'simplicity', *drîvalte* 'trinity', *gewaltec* 'violent'. With or without late stage correction, though, the limited lexical exceptionality we find in the Old High German data is at most a minor problem for a phonological account of umlaut failure.

6. Conclusion. Umlaut failure in Old High German and across Germanic generally is, we conclude, clearly phonological, driven by one principle: the greater the phonetic difference between target and trigger, the more likely umlaut is. Less difference correlates then directly with umlaut failure. Even the pattern of exceptionality finds parallels in other vowel harmony systems. Less principled analogical accounts must be dismissed. The hierarchy of umlaut failure reflects the likely chronological unfolding of umlaut in Upper German, where forms showing the most failure were the last reached by umlaut (shown in 9, below).

Across the board, it is the umlautless residues, evidence of where and how umlaut failed, that provide the key to understanding the complex unfolding of umlaut over time and space. In primary umlaut, a process limited to a single short vowel, intervening vocalized liquids systematically removed many umlautable forms. The last forms to umlaut were those showing the broadest sharing of features, notably between vowels and in the secondary place features of intervening consonants. The process of Old High German umlaut—from the early days of primary umlaut with LC/*h*C blocking to the last stages of /u/ blocked by geminate velars—shows remarkable phonological coherence, viewed in the right perspective, and a steady progression, shown in 9.

(9) The generalization of umlaut
1. Primary: only short /a/ target; intervening V-place as target coda blocks: /a/, but not *a*LC, *ah*C
2. Nonprimary:
   a. long and short targets, without former blocking environments: /a:/, /o/, etc.
   b. increasingly higher vowels, to /u/
   c. within the /u/ cases, geminates and place-sharing target codas block: Everything but *-ukxi*, etc.
3. Consistent umlaut.

Over the vast majority of the German-speaking territory, umlaut ultimately generalized across *all* environments, reaching stage 3. The few southern relict areas retaining umlautlessness, of type 2c, shed light on the last steps in umlaut's evolution. Even within those relict areas, a hierarchy is apparent, with geminate velars at the top, followed by labial and coronal geminates.

We started with the simple task of showing regular phonological development to be the source of *Umlautfeindlichkeit* in Upper German, but something broader has emerged: the many and variegated umlaut phenomena found across Germanic *all* operate and fail to operate in accordance with one basic principle of assimilation.

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